RETRIEVAL AUGMENTED GENERATION

Presented by
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Hallucinations in Large Language Models (LLMs)

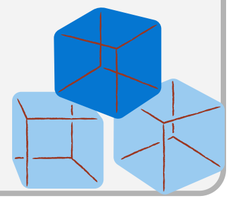
What is Hallucination?

- LLMs generate fluent, confident, but factually incorrect or fabricated information



Example of Hallucination?

- User: Who wrote the famous book The Secret Life of Algorithms?
- Response: The book The Secret Life of Algorithms was written by Dr. John Peterson
- Reality: No such book exists. The LLM completely invented the title, author



Why Do LLMs Hallucinate??

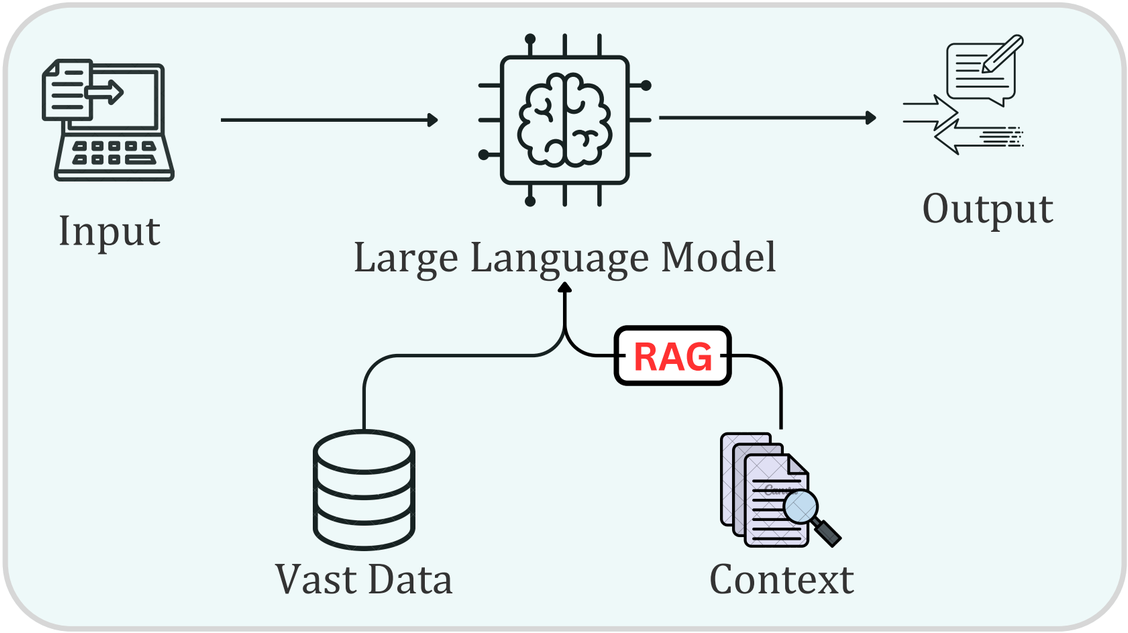
- The model may lack relevant or up-to-date information.
- Unclear questions can lead to unsupported assumptions.



How RAG solves these problems?

RAG (Retrieval-Augmented Generation) is a technique used to provide relevant external context to a Large Language Model by retrieving information from documents or databases before generating an answer.

- **Retrieve:** The AI first searches for the most relevant information from documents, websites, or databases.
- **Augment:** It adds that real information to what it already knows.
- **Generate:** Then it creates a more accurate and factual answer using both the searched info and its own knowledge.



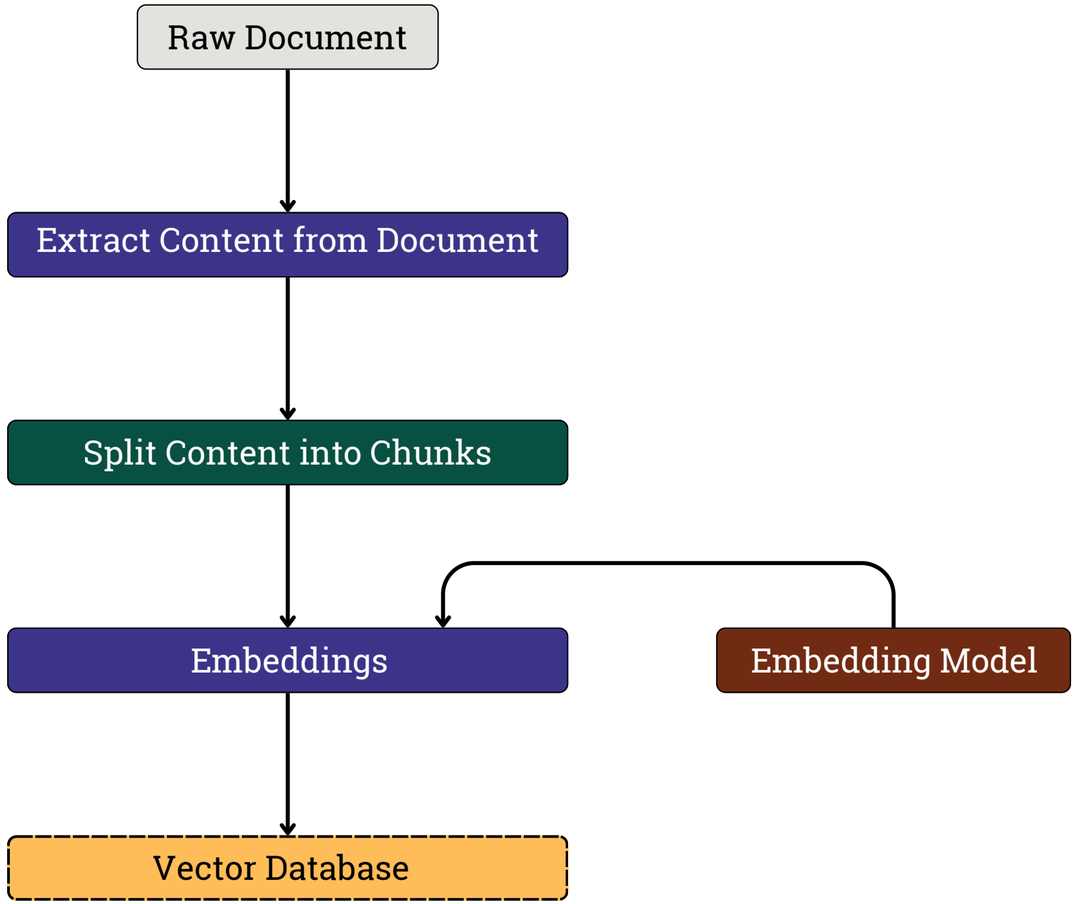
Phase 1: Indexing

What is Indexing?

- Prepare and store knowledge before users ask questions.

What is Output from Indexing?

- Searchable knowledge base containing document chunks and their vector embeddings.



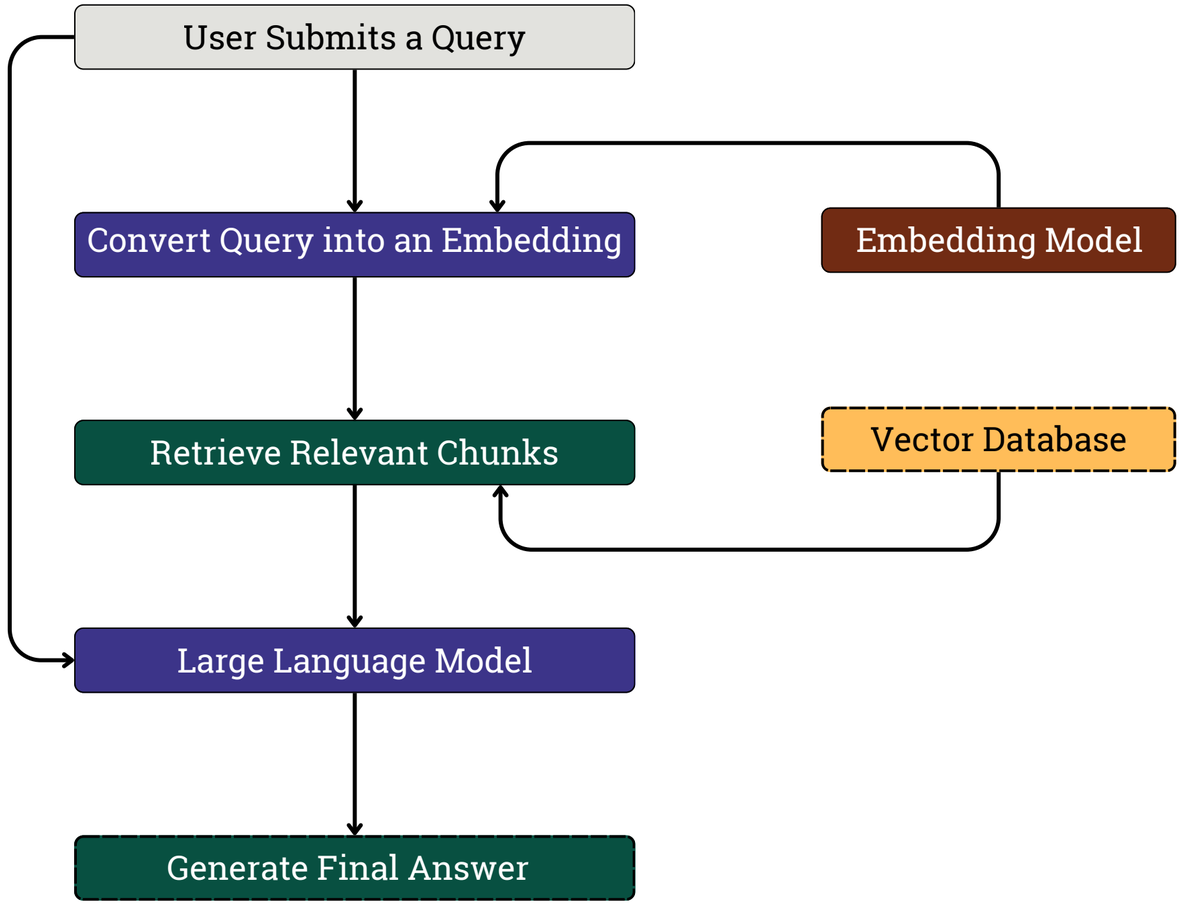
Phase 2: Retrieval & Generation

What is Retrieval & Generation?

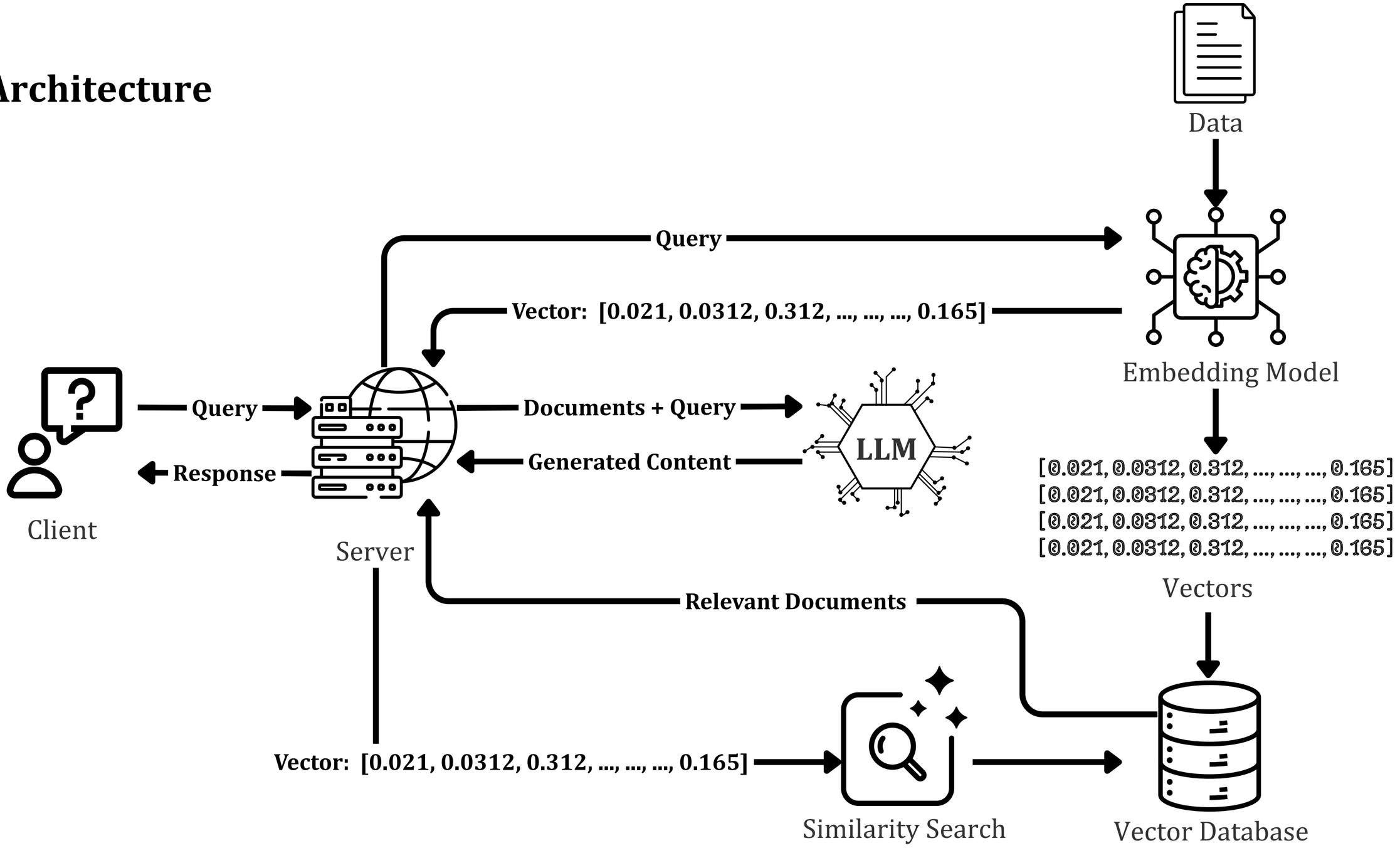
- Find relevant information and answer the user's question.

What is Output from Indexing?

- An answer generated using the retrieved context.

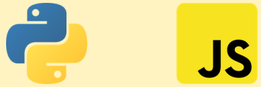


Architecture

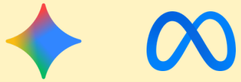


Technologies used for development - RAG Pipeline

Programming Language — *Python, Javascript*



Embedding Model — *Gemini, Llama, Opensource*



AI Framework — *Langchain, LlamaIndex*



UI Development — *Nextjs, TailwindCSS*



Vector Database — *ChromaDB, PineCone, pgVector*



Version Control — *GitHub, Gitlab*



Large Language Model — *Gemini, Gemma, Llama*



Testing Framework — *Playwright, Selenium*



AI Platforms — *Hugging Face, Groq AI*



Integration — *Axios, Fetch API*

AXIOS

Deployment — *AWS, GCP, Render, Vercel*



Where is RAG used?

Customer Support & Chatbots

Provides accurate, context-aware answers by retrieving FAQs, tickets, and product docs.

Finance & Banking

Retrieves regulatory rules, account info, and market data for accurate financial guidance.

Enterprise & Internal Search

Helps employees find internal documents and summaries quickly using company data.



Legal & Compliance

Summarizes contracts, statutes, and case law for faster legal research and drafting.

Healthcare & Life Sciences

Supports clinicians with patient history, research, and guidelines for evidence-based decisions.

Content Creation & Marketing

Generates marketing copy and content grounded in brand guidelines and documents.

Thank You!



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